

Rahini Raneru

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Education

B.Tech in Computer Science Engineering

2024 – 2028

Manipal University Jaipur, Rajasthan, India

CGPA: 9.00 / 10 (Dean's List, Semester 2 & 4)

Relevant Coursework: Data Structures and Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Operating System, Design and Analysis of Algorithms

Class 12

2019 – 2024

Delhi Public School Nacharam, Hyderabad, India

Percentage: 91

Technical Projects

GoalCast - Player Goal Probability Predictor | [GitHub]

- Developed a CatBoost-based football player goal prediction system trained on 1.3M+ player-match records; addressed class imbalance, feature leakage, and missing value imputation across a large, real-world dataset.
- Engineered domain-specific features spanning expected goals (xG), shooting metrics, market valuation, positional role, and international experience; achieved Average Precision of 47.74 (competition best: 53.00), within 10% of the competition-leading score.
- Built an end-to-end ML pipeline covering data ingestion, preprocessing, feature engineering, model training, and batch inference on held-out match data.
- **Technologies:** Python, Pandas, NumPy, CatBoost, Scikit-learn

RentMan: Rental Management System

- Independently designed and built a full-stack rental management system with a normalized MySQL schema covering six entities: Owners, Tenants, Properties, Rental Agreements, Payments, and Issue Tracking.
- Developed RESTful API endpoints using Node.js and Express.js for CRUD operations across all entities; used Sequelize ORM for model associations and automated query generation.
- Built a dynamic frontend using HTML, CSS, and JavaScript with Fetch API integration to retrieve and display backend data without page reloads.
- Validated all API endpoints using Postman, testing request handling, JSON responses, and error cases across the full data flow.
- **Technologies:** Node.js, Express.js, MySQL, Sequelize ORM, HTML, CSS, JavaScript, Postman

Sleep Stage Classification from EEG

- Built an ML pipeline to classify 5 sleep stages from raw EEG signals using the PhysioNet Sleep-EDF dataset (~197 hours of annotated recordings).
- Preprocessed raw EEG data using MNE-Python (signal filtering, epoching, and artifact handling) to prepare recordings for feature extraction.
- Extracted frequency-domain features (delta, theta, alpha, beta band powers) using NumPy and SciPy; trained a Random Forest classifier achieving 82% accuracy and 0.71 macro F1-score on held-out epochs across class-imbalanced sleep stage data.
- **Technologies:** Python, MNE-Python, NumPy, SciPy, Pandas, Scikit-learn

NotifyMail – AI-Powered Email Triage System | [GitHub]

- Built a production-deployed email intelligence pipeline that monitors 2 Gmail accounts, classifies incoming emails using Groq's Llama 3.1 8B, and delivers structured Discord notifications — eliminating missed internship and hackathon deadlines.
- Engineered a multi-stage n8n workflow with OAuth2 Gmail integration, LLM-based JSON classification with prompt-engineered inclusion rules for hackathons and opportunities, and conditional routing via parsed AI output.
- Deployed on Render with Neon PostgreSQL for workflow persistence and GitHub Actions as a cron-based web-hook trigger; resolved ephemeral storage, encryption key persistence, and OAuth credential lifecycle issues in production.
- **Technologies:** n8n, Groq API (Llama 3.1 8B), Gmail API (OAuth2), Docker, Render, Neon PostgreSQL, GitHub Actions, JavaScript

Technical Skills

- **Programming:** Python, C, Java
- **Libraries/Data:** NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, PyTorch
- **Web:** HTML, CSS, JavaScript, TypeScript, Node.js, Express.js
- **Tools:** Git, GitHub, LaTeX, Overleaf, Postman, Claude Code
- **Automation:** n8n
- **Databases:** MySQL, PostgreSQL

Technical Experience

Webmaster in IEEE Computational Intelligence Society

2025 – 2026

Manipal University Jaipur

- Maintained and updated the chapter website with event information, blog posts, and crew profiles using HTML, TypeScript, CSS, and JavaScript.
- Enhanced site accessibility and responsiveness by reorganizing the navigation bar and improving section layouts for better usability.
- Co-organized **Prompt401**, a prompt engineering workshop and competition attended by 40+ participants, managed participant registrations, coordinated group communications, and supported social media announcements.

Open Source Contributor

2025

GirlScript Summer of Code

- Contributed to a browser-based video editor/enhancer project - submitted 2 pull requests with 1 merged, adding new UI sections and improving the existing interface.
- Practiced Git-based collaborative workflows including forking, branching, and pull request submissions within a multi-contributor codebase.

Research Experience

TrustA2A - Research Project

- Surveyed 15+ papers on trust mechanisms in multi-agent systems, mapping existing approaches to agent-to-agent reliability evaluation.
- Identified open research gaps in trust modeling for autonomous communication frameworks and scoped a prototype architecture for trust scoring.
- Implementing a Python-based simulation to model trust propagation between agents, currently in early development.
- Produced structured research documentation in LaTeX covering literature synthesis, gap analysis, and proposed methodology.

Certifications

- **Introduction to Artificial Intelligence** – Coursera
- **Database Design** – Oracle Academy
- **Database Foundations** – Oracle Academy
- **Programming, Data Structures and Algorithms using Python** – NPTEL
- **Design and Analysis of Algorithms** – NPTEL
- **Hacks 9.0 Hackathon Finalist** – ACM MUJ
- **Hacks 10.0 Hackathon Finalist** – ACM MUJ

Leadership & Extracurriculars

Lance Corporal—National Cadet Corps

Delhi Public School, Nacharam

- Achieved rank of Lance Corporal through structured NCC training focused on discipline, leadership and teamwork.

Rapporteur—Epistemia Science & Technology MUN

Delhi Public School, Nacharam

- Documented and synthesized technical discussions and research-oriented debates during committee sessions.
- Presented structured reports summarizing the policy discussions and scientific perspectives presented by the delegates.